

CSDA5110

Fall 1 2024

Homework 1 due 5:00pm Friday, August 30

Read Chapters 1 through 4 in R for Everyone: Advanced Analytics and Graphics and then complete the following assignment.

The following assignment can be done using R markdown. You knit the document as word document. You should save the document as pdf and upload a single pdf document in the dropbox. This document should contain all the code and output too.

FIRST - INSTALL R | Follow the book's guidelines for setup

1. Download and install R version 4.4.1 from the Cran R repository for the appropriate OS your laptop or desktop uses (Windows will download the .exe file, Mac OS will be a .pkg file). The link to the Cran R repository is: [The Comprehensive R Archive Network \(r-project.org\)](https://cran.r-project.org/)
2. **Confirm you have installed the latest version of R by typing "R.version" into the R console window and paste a screenshot of the R Console into a Word document.**

SECOND - INSTALL R STUDIO | Follow the book's guidelines for setup

1. Download and install the **FREE** version of RStudio from the following link: [RStudio Desktop - Posit](https://www.rstudio.com/products/rstudio/download/#download)
2. **Confirm you have installed RStudio by pasting a screenshot of the R Studio IDE into the same Word document.**

THIRD - WRITE THE CODE

1. Create four variables that are assigned a numeric data type, a character data type, a logical data type, and a date data type. Each variable should contain a different data type than the others.
 - a. Validate the type of data contained in the variable using the appropriate function that validates the class of the variable. (**HINT**: Read section 4.3 in the book)
2. Create a variable called "monthsOfYEAR" that contains a vector of the names of the days of the week.
3. Create a variable called "dataVec1" that contains a vector of numbers that range from 200 to 220.
4. Create a variable called " dataVec2" that contains a vector of numbers that range from 40 to 60.
5. Create a variable called " datacombVec1" that adds the vectors of " dataVec1" and " dataVec2" into the new variable " datacombVec1".
6. Create a variable called " datacombVec2" that multiplies the vectors of " dataVec1" and " dataVec2" into the new variable " datacombVec2".
7. What's the difference between NULL and NA? How can NAs be resolved when using the sum command?
8. Write out the code that you would type in the RStudio script editor to install and load the package dplyr.